# MES's Abasaheb Garware College, Pune
# M.Sc. (Computer Science) – II
# CS-604-MJP Lab course based on CS-602 and CS-603

## Full Stack Development- II Practical Assignment Questions

1. Create a basic React application that displays your name, a short bio, and an image.

2. Design a functional component to display a list of 5 favorite movies using JSX.

3. Demonstrate the concept of virtual DOM by comparing traditional DOM vs React rendering through a counter example.

4. Create reusable components like Header, Footer, and Card, and use them in a homepage layout.

5. Build a counter app using useState to increment and decrement a number.

6. Create a functional component using useEffect() that fetches and displays data from an API (e.g., users list).

7. Demonstrate the use of useReducer() hook by building a to-do list app with add/delete functionality.

8. Create a custom hook to manage form input fields with validation (e.g., email or password).

9. Build a theme toggler app using useContext() hook to switch between dark and light modes.

10. Create a login form with event handlers for input and submit events. Display the entered values in the console.

11. Demonstrate synthetic events by creating a button that logs click and double-click events differently.

12. Create a multi-page React app using react-router-dom with Home, About, and Contact pages.

13. Pass parameters through routes and display the dynamic data (e.g., /user/:id should show user details).

14. Create a protected route that only shows a page if the user is authenticated (mock authentication).

15. Set up Redux in a React app and create a simple counter using actions and reducers.

16. Build a shopping cart app where Redux manages the state of cart items (add, remove, quantity).

17. Design a login system with mock authentication using hardcoded user data and simulate token-based login using JWT (can be mocked).

18. Build a notes management app using Redux Toolkit with createSlice() to manage state.

19. Use RTK Query to fetch posts from a fake API (like JSONPlaceholder) and display them with loading/error states.

20. Demonstrate the migration to Redux Toolkit by converting a traditional Redux app into one using Redux Toolkit and compare both.

21. Create React Application to take input from user (number, string). check num is Even or odd and also check string is palindrome or not. display appropriate message.

22. Create React Application to create calculator.

23. Create React Application which contains toggle button when button is clicked, text should toggle betn ON & OFF, Background color of text should also changed, visually reflect the state.

24. Create React Application to check number is valid or not, also calculate sum of digit of number

25. Create React Application, which will perform string operations. (uppercase, lowercase, Reverse)